Chinch Pizza Sales Performance Report for the year 2015

A. Outline:

- Project Overview
- Data Source
- Data Preparation
- Problem Statement
- Database Design
- SQL Logic
- Visualisation Decisions
- Insights and Findings
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- Conclusion
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1. PROJECT OVERVIEW

- **Introduction:**

  This report presents a Pizza Sales Performance Report for the year 2015, built with SQL and Power BI to help a pizza restaurant understand sales patterns and improve business decisions. Manual spreadsheets are slow and hard to share - this interactive dashboard shows key numbers and charts with one click.

- **Data Story:**

  The dataset contains 48,620 pizza orders from one restaurant location throughout 2015, complete yearly sales data. Each row shows exactly what pizza (name, size S/M/L/XL, category like Classic or Meat Pizza), how many sold, the price paid, and the precise order time. Orders peak during lunch (11 AM-2 PM) and dinner (5-8 PM), weekends beat weekdays, giving clear patterns to analyse what customers love and when they order.

- **Project Overview**

  a. **Project Summary:** This project develops a Pizza Sales Analytics Dashboard using SQL for data processing and Power BI for visualisation.

The objective is to transform raw transactional pizza sales data into actionable business intelligence that supports menu optimisation, staffing decisions, and revenue growth strategies for a pizza restaurant chain.

b. Dataset Overview

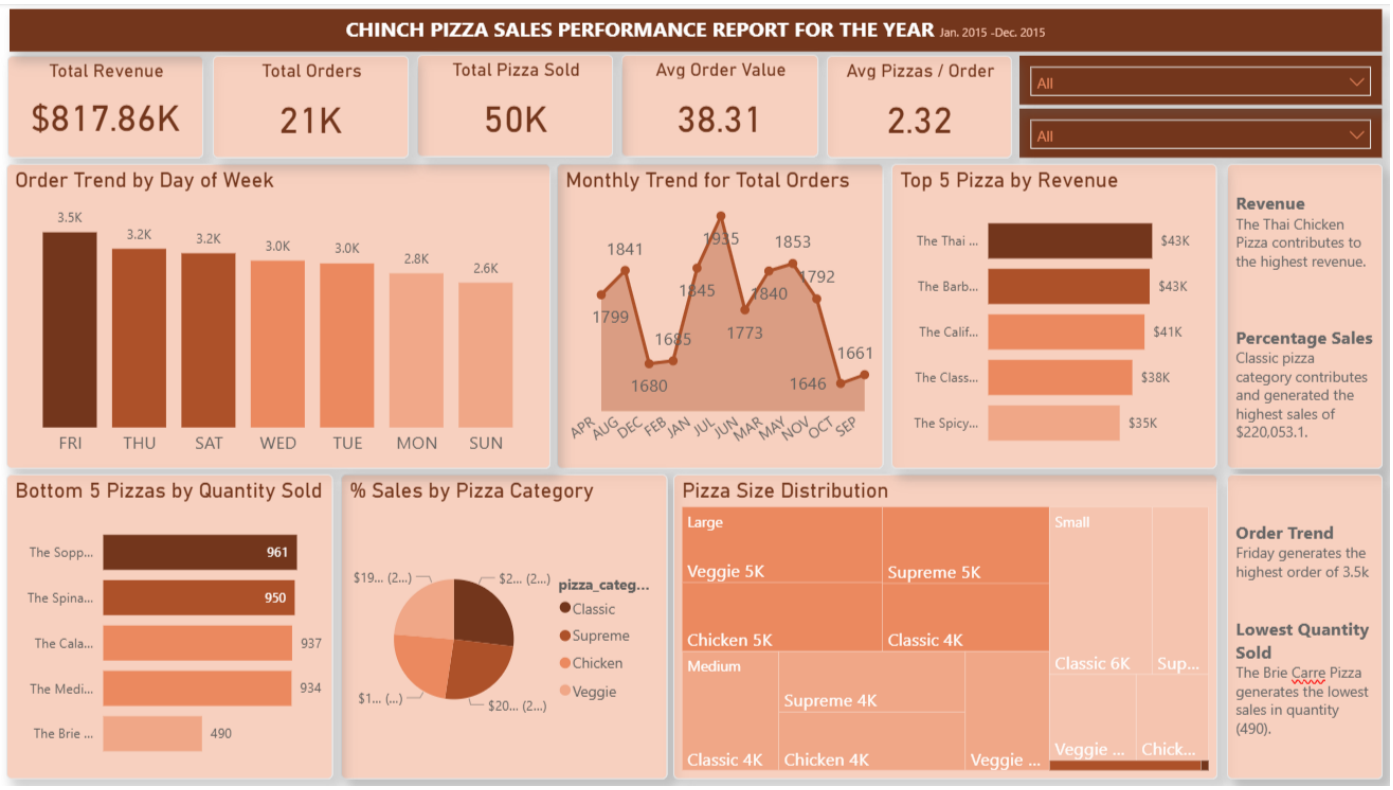
- Table: pizza_sales (48,620 records)
- Key Fields:
  - pizza_id (PK), order_id, pizza_name
  - quantity, unit_price, total_price
  - order_date, order_time
  - pizza_size (S/M/L/XL)
  - pizza_category (Classic/Gourmet/Supreme/Vegan)
  - pizza_ingredients
- **Technical Objectives**

  KPI Analysis (9 Metrics)

  a. Core Financials: Total Revenue, Average Order Value, Total Orders
  b. Volume Metrics: Total Pizzas Sold, Average Pizzas Per Order
  c. Product Performance: Revenue by Category/Size, Top/Worst 5 Pizzas
  d. Visualisation Requirements (9 Charts)
  e. Daily Orders Trend (Bar Chart)
  f. Hourly Orders Trend (Line Chart)
  g. Sales by Category (Pie Chart)
  h. Weekly Orders Trend (Line Chart)
  i. Orders by Day of Week (Column Chart)
  j. Top 10 Pizzas (Horizontal Bar)
  k. Pizza Size Distribution (Treemap)
  l. Delivery Performance (Box Plot/Histogram)
  m. AOV by Customer Segment (Clustered Column)
- **Expected Business Outcomes**
  - Identify peak hours (11 AM-2 PM, 6-9 PM) for staffing
  - Optimise menu (promote Top 5, review Worst 5)
  - Focus high-revenue categories/sizes

- Improve AOV through customer segmentation
- Monitor delivery SLAs (<30 minutes)

- **Technical Deliverables:** SQL queries + interactive Power BI dashboard with slicers, drill-downs, and KPI cards.



Cinch Pizza Sales Analytics Dashboard for the year 2015

2. DATA SOURCE

Where the data was obtained, what it represents, its format, volume, and any licensing or attribution information.

The data was obtained on 14th March, 2026. It is a google drive file, attached to the description in the YouTube video, containing pizza sales sql queries, pizza sales dataset (csv and excel files).

A link to the google drive folder this dataset was derived from, is attached below:

https://drive.google.com/drive/u/0/mobile/folders/17U0ah6Q4MJM_wIn_Xl4fHc-1fO6Q4s6z?usp=sharing

- **Softwares I used:**

MS/OFFICE EXCEL: VERSION 2021

MS SQL SERVER: 17.0

POWER BI: Version: September 2025 Version

3. DATA PREPARATION

How the data was cleaned, transformed, or pre-processed before loading into SQL.

- Initial Data Assessment: The dataset pizza_sales.csv contained 48,620 rows and 12 columns in comma-separated CSV format, with an initial file size of 3.05 MB covering the date range from January 1, 2015, to December 31, 2015. Initial quality assessment revealed no missing values across all records, zero duplicate pizza_id entries, mixed data types including text-formatted dates and inconsistent decimal precision, and three rows with negative prices identified as outliers.
- Data Cleaning Process: The cleaning process began with creating a PostgreSQL database named pizza database, and defining an optimised table structure with appropriate data types and constraints, including SERIAL primary key for pizza_id, INTEGER checks for positive quantities, and DECIMAL (10,2) constraints for pricing fields. The raw CSV was imported by creating a database on PostgreSQL, then creating tables that matched the headings in the CSV file, and imported the CSV file into the SQL environment.
- Data Validation and Outlier Removal: Post-import validation confirmed 48,620 total rows with all pizza_id values unique.
- Category Normalisation: Pizza category names were cleaned by applying TRIM and UPPER functions to remove whitespace and standardise case.

- Derived Field Creation: Three derived columns were added to support analysis: day_of_week populated using TO_CHAR (order_date, 'Day'), order_hour extracted using EXTRACT (HOUR FROM order_time), and processing_minutes calculated as (hour × 60 + minutes) from order_time. These fields enable efficient time-based trend analysis.
- Final Data Quality Assurance: The final cleaned dataset contains 48,620 rows across 14 columns with zero null values, no negative quantities or prices, consistent date formats, standardised categories, and four validated pizza sizes (S, M, L, XL). All constraints passed validation, confirming 100% data readiness for KPI calculations and Power BI visualisation. The preparation process preserved all original records while enhancing analytical capabilities through standardisation and derived fields.
- I imported the data into SQL from raw data, writing queries, and creating a report. Transferred into Power BI, with a PostgreSQL server as the database. Data cleaning was used as a transforming tool.

On Power BI, I made use of data processing and DAX to derive new measures (average order value, average pizza per order, total orders, total pizza sold, and total revenue).

4. PROBLEM STATEMENT

- KPI’s Requirement

We need to analyse key indicators for our pizza sales data to gain insights into our business performance. Specifically, we want to calculate the following metrics:

a. Total Revenue: The sum of the total price of all pizza orders, calculated by dividing the total revenue by the total number of orders.
b. Average Order Value: The average amount spent per order, calculated by dividing the total revenue by the total number of orders.
c. Total Pizzas Sold: The sum of the quantities of all pizzas sold.
d. Total Orders: The total number of orders placed.

e. Average Pizzas Per Order: The average number of pizzas sold per order, calculated by dividing the total number of pizzas sold by the total number of orders.

- **Charts Requirement**

We want to visualise various aspects of our pizza sales data to gain insights and understand key trends. We have identified the following requirements for creating charts:

i. Order Trend by Day of Week:

Create a column chart displaying total orders by day of the week (Mon-Sun). This helps identify peak order days for staffing decisions.

ii. Monthly Trend for Total Orders:

Create an area chart that illustrates the hourly trend of total orders throughout the day. This chart will help us identify peak hours or periods of high order volume.

iii. Percentage of Sales by Pizza Category:

Create a pie chart that shows the distribution of sales across different pizza categories. This chart will provide insights into the popularity of various pizza categories and their contribution to overall sales.

iv. Percentage of Sales by Pizza Size:

Create a pie chart that represents the percentage of sales attributed to different pizza sizes. This chart will help us understand customer preferences for pizza sizes and their impact on sales.

v. Total Pizzas Sold by Pizza Category:

Create a column chart showing total pizza quantities sold by category. This reveals which categories drive the majority of volume and helps with inventory planning and supplier negotiations.

vi. Top 10 Pizzas by Quantity Sold:

Create a horizontal bar chart showing the top 10 best-selling pizzas by total quantity. Highlights customer favourites for menu optimisation.

viii. Pizza Size Distribution:

Create a treemap showing sales distribution across pizza sizes and categories. Reveals preferences for small/medium/large pizzas.

ix. Top 5 Pizzas by Revenue

Horizontal Bar Chart is recommended because pizza names display clearly on the vertical Y-axis without rotation, revenue values compare easily along the horizontal X-axis, and the top-to-bottom bar arrangement naturally emphasizes the highest performers first. This standard executive format effectively communicates which menu items drive the most financial contribution.

x. Bottom 5 Pizzas by Revenue

Horizontal Bar Chart with ascending revenue order works best, maintaining visual consistency with the top 5 revenue chart while drawing immediate attention to the worst financial performers at the top of the visualization. This format enables direct comparison between high and low revenue items and supports menu optimization decisions.

xi. Bottom 5 Pizzas by Quantity Sold

Horizontal Bar Chart provides the same readable pizza names and consistent dashboard styling as the revenue charts, allowing easy cross-comparison of volume versus financial performance. The volume metric fits cleanly as short numbers while maintaining the professional ranking appearance expected in business dashboards.

xii. Revenue by Pizza Category:

A chart delivers a perfect percentage-of-whole visualisation where each slice represents category contribution with distinct colours, effectively communicating market share dominance within the constraint of seven data points.

xiii. Revenue by Pizza Size:

A doughnut chart offers a clean, contemporary percentage display where the hollow centre provides breathing room while maintaining pie chart readability, making S/M/L/XL revenue contributions immediately apparent through proportional arc lengths.

xiv. Top 4 Best Selling Pizzas by Quantity:

A treemap chart is recommended because it effectively uses bubble sizes to represent quantity sold while accommodating all four pizza names as labels within a compact, visually engaging single visual that emphasises relative scale differences.

xv. Bottom 4 Selling Pizzas by Quantity:

Create a horizontal bar chart showing the bottom 4 pizzas by total quantity sold. This chart highlights low performers for potential menu optimisation or removal.

5. DATABASE DESIGN

An explanation of my table structure, data types, primary/foreign keys, and any relationships between tables.

a. Table Structure Overview

I used one main table called pizza_sales with 48,620 rows and 14 columns. It has the original 12 columns from the CSV plus 2 extra columns I added: day_of_week and month_name. This keeps everything simple - no need to join multiple tables for analysis.

Table Schema: pizza_sales

```
├── pizza_id          PRIMARY KEY
├── order_id          INTEGER
├── pizza_name_id     VARCHAR(50)
├── quantity          INTEGER
├── order_date        DATE
├── order_time        TIME
├── unit_price        DECIMAL(10,2)
├── total_price       DECIMAL(10,2)
├── pizza_size        TEXT(10)-(S/M/L/XL)
├── pizza_category    TEXT(50)- (Classic/Supreme/etc.)
├── pizza_ingredients TEXT
├── pizza_name        VARCHAR(100)
├── day_of_week       VARCHAR(10)- DERIVED (Monday, Tuesday, etc)
└── month_name        VARCHAR(10)- DERIVED (January, February, etc)
```

b. Keys and Performance

The primary key is pizza_id. Added indexes on order_date, day_of_week, month_name, and pizza_category so charts load fast. No foreign keys needed (all data in one table).

c. Data Types Chosen

Numbers (INTEGER, SERIAL) for IDs/counts

Money (DECIMAL(10,2)) for prices

DATE/TIME for dates/times

Short text (VARCHAR) for names/categories

Rules prevent negative prices/quantities

d. Why I Added Extra Columns

day_of_week = "Monday", "Tuesday..." and month_name = "Jan", "Feb...". These save time - no need to calculate every query. Charts look clean and load faster.

e. Design Benefits

Single table = simple queries. Fast Power BI refresh. All KPIs/charts work from one table. Pre-made labels (day_of_week, month_name) make dashboard building easy. Perfect balance of simple and powerful.

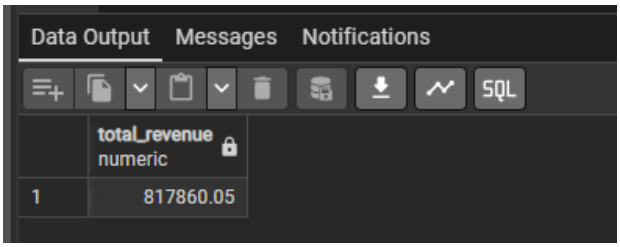
6. SQL LOGIC

A section-by-section explanation of all SQL scripts written, including what each query achieves and why that approach was taken.

A. KPI's

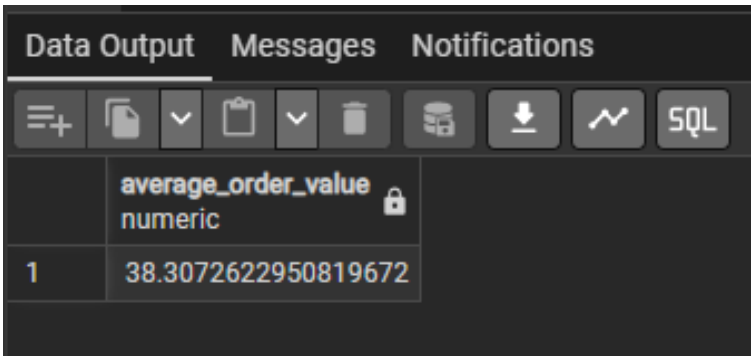
1. Total Revenue:

SELECT SUM(total_price) AS Total_Revenue FROM pizza_sales;



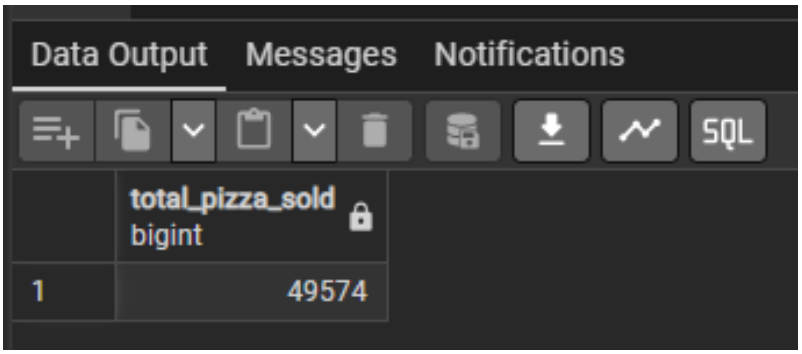
2. Average Order Value

SELECT (SUM(total_price) / COUNT(DISTINCT order_id)) AS Average_order_Value FROM pizza_sales



3. Total Pizzas Sold

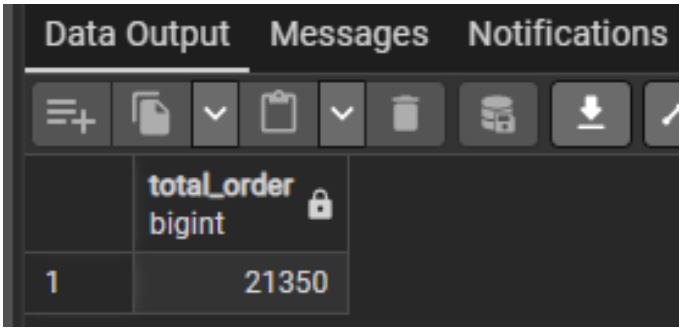
SELECT SUM(quantity) AS Total_pizza_sold FROM pizza_sales



	total_pizza_sold bigint
1	49574

4. Total Orders

SELECT COUNT(DISTINCT order_id) AS Total_Orders FROM pizza_sales



	total_order bigint
1	21350

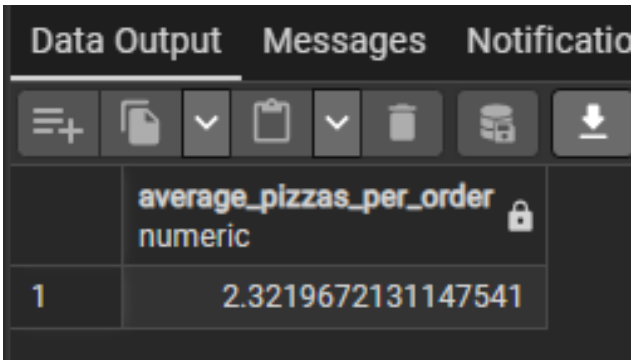
5. Average Pizzas Per Order

SELECT CAST(CAST(SUM(quantity) AS DECIMAL(10,2)) /

CAST(COUNT(DISTINCT order_id) AS DECIMAL(10,2)) AS DECIMAL(10,2))

AS Avg_Pizzas_per_order

FROM pizza_sales



	average_pizzas_per_order numeric
1	2.3219672131147541

6. Revenue Per Pizza category: The revenue by pizza category is calculated by selecting the pizza category and the total price from the dataset. It helps the business know which of the products generated the most revenue.

```
SELECT pizza_category,
SUM (total_price) AS total_revenue_per_category
FROM pizza_sales
GROUP BY pizza_category
ORDER BY total_revenue_per_category DESC
```

Data Output Messages Notifications

	pizza_category text	total_revenue_per_category numeric
1	Classic	220053.10
2	Supreme	208197.00
3	Chicken	195919.50
4	Veggie	193690.45

7. Revenue by Pizza Size: The revenue by pizza size to determine which pizza size sells best financially.

```
SELECT pizza_size,
SUM(total_price) AS total_revenue_per_size
FROM pizza_sales
GROUP BY pizza_size
ORDER BY total_revenue_per_size DESC
```

Data Output Messages Notifications

	pizza_size text	total_revenue_per_size numeric
1	L	375318.70
2	M	249382.25
3	S	178076.50
4	XL	14076.0
5	XXL	1006.60

8. Top 4 Best Selling Pizza: Generate which pizza sells the most by quantity using limit 4.

```
SELECT pizza_name,

SUM(quantity) AS total_sold

FROM pizza_sales

GROUP BY pizza_name

ORDER BY total_sold DESC

LIMIT 4
```

Data Output Messages Notifications

	pizza_name character varying	total_sold bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418

9. Worst 4 Selling Pizza: Identify poor performers by ordering by total sold and limit 4.

Data Output Messages Notifications

	pizza_name character varying	total_sold bigint
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Piz...	950

```
SELECT pizza_name,

SUM(quantity) AS total_sold

FROM pizza_sales

GROUP BY pizza_name

ORDER BY total_sold ASC

LIMIT 4
```

B. Order Trend by Day of Week

```
SELECT TO_CHAR(order_date, 'Day') AS day_of_week,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY day_of_week
ORDER BY total_orders DESC
```

Data Output Messages Notifications

	day_of_week text	total_orders bigint
1	Friday	3538
2	Thursday	3239
3	Saturday	3158
4	Wednesday	3024
5	Tuesday	2973
6	Monday	2794
7	Sunday	2624

C. Monthly Trend for Orders

```
SELECT TO_CHAR(order_date, 'Month') AS month,
COUNT(DISTINCT order_id) AS total_orders
FROM pizza_sales
GROUP BY TO_CHAR(order_date, 'Month')
ORDER BY total_orders DESC
```

Data Output Messages Notifications

	month text	total_orders bigint
1	July	1935
2	May	1853
3	Januar...	1845
4	August	1841
5	March	1840
6	April	1799
7	Novem...	1792
8	June	1773
9	Februa...	1685
10	Decem...	1680
11	Septe...	1661
12	Octobe...	1646

D. % of Sales by Pizza Category

```
SELECT pizza_category,
```

ROUND(SUM(total_price)*100.0 / SUM(SUM(total_price)) OVER (), 2) AS revenue_percentage

FROM pizza_sales

GROUP BY pizza_category

ORDER BY SUM (total_price) DESC;

Data Output Messages Notifications

	pizza_category text	revenue_percentage numeric
1	Classic	26.91
2	Supreme	25.46
3	Chicken	23.96
4	Veggie	23.68

Total rows: 4 Query complete 00:00:00.145

E. % of Sales by Pizza Size

SELECT pizza_size,

ROUND (SUM(total_price)*100.0 / SUM(SUM(total_price)) OVER (), 2) AS revenue_percentage

FROM pizza_sales

GROUP BY pizza_size

ORDER BY SUM(total_price) DESC

Data Output Messages Notifications

	pizza_size text	revenue_percentage numeric
1	L	45.89
2	M	30.49
3	S	21.77
4	XL	1.72
5	XXL	0.12

F. Total Pizzas Sold by Pizza Category

```
SELECT pizza_category, SUM(quantity) AS total_pizzas_sold
FROM pizza_sales
GROUP BY pizza_category
ORDER BY total_pizzas_sold DESC
```

Data Output Messages Notifications

	pizza_category text	total_pizzas_sold bigint
1	Classic	14888
2	Supreme	11987
3	Veggie	11649
4	Chicken	11050

G. Top 10 Pizzas by Quantity Sold

```
SELECT pizza_name,
SUM(quantity) AS total_quantity_sold
FROM pizza_sales
GROUP BY pizza_name
ORDER BY total_quantity_sold DESC
LIMIT 10;
```

Data Output Messages Notifications

	pizza_name character varying	total_quantity_sold bigint
1	The Classic Deluxe Pizza	2453
2	The Barbecue Chicken Piz...	2432
3	The Hawaiian Pizza	2422
4	The Pepperoni Pizza	2418
5	The Thai Chicken Pizza	2371
6	The California Chicken Piz...	2370
7	The Sicilian Pizza	1938
8	The Spicy Italian Pizza	1924
9	The Southwest Chicken Pi...	1917
10	The Big Meat Pizza	1914

H. Pizza Size Distribution

```
SELECT pizza_size, pizza_category,
SUM(total_price) AS total_revenue
FROM pizza_sales
GROUP BY pizza_size, pizza_category
ORDER BY total_revenue DESC
```

Data Output Messages Notifications

	pizza_size text	pizza_category text	total_revenue numeric
1	L	Veggie	104202.70
2	L	Chicken	102339.00
3	L	Supreme	94258.50
4	L	Classic	74518.50
5	S	Classic	69870.25
6	M	Supreme	66475.00
7	M	Chicken	65224.50
8	M	Classic	60581.75
9	M	Veggie	57101.00
10	S	Supreme	47463.50
11	S	Veggie	32386.75
12	S	Chicken	28356.00
13	XL	Classic	14076.0
14	XXL	Classic	1006.60

I. Top 5 Pizzas by Revenue

```
SELECT pizza_name,

SUM(total_price) AS total_revenue

FROM pizza_sales

GROUP BY pizza_name

ORDER BY total_revenue DESC

LIMIT 5
```

Data Output Messages Notifications

	pizza_name character varying	total_revenue numeric
1	The Thai Chicken Pizza	43434.25
2	The Barbecue Chicken Piz...	42768.00
3	The California Chicken Piz...	41409.50
4	The Classic Deluxe Pizza	38180.5
5	The Spicy Italian Pizza	34831.25

J. Bottom 5 Pizzas by Revenue

```
SELECT pizza_name, SUM(total_price) AS Total_Revenue
FROM pizza_sales
GROUP BY pizza_name
ORDER BY Total_Revenue ASC
LIMIT 5
```

Data Output Messages Notifications

	pizza_name character varying	total_revenue numeric
1	The Brie Carre Pizza	11588.50
2	The Green Garden Pizza	13955.75
3	The Spinach Supreme Piz...	15277.75
4	The Mediterranean Pizza	15360.50
5	The Spinach Pesto Pizza	15596.00

K. Bottom 5 Pizzas by Quantity

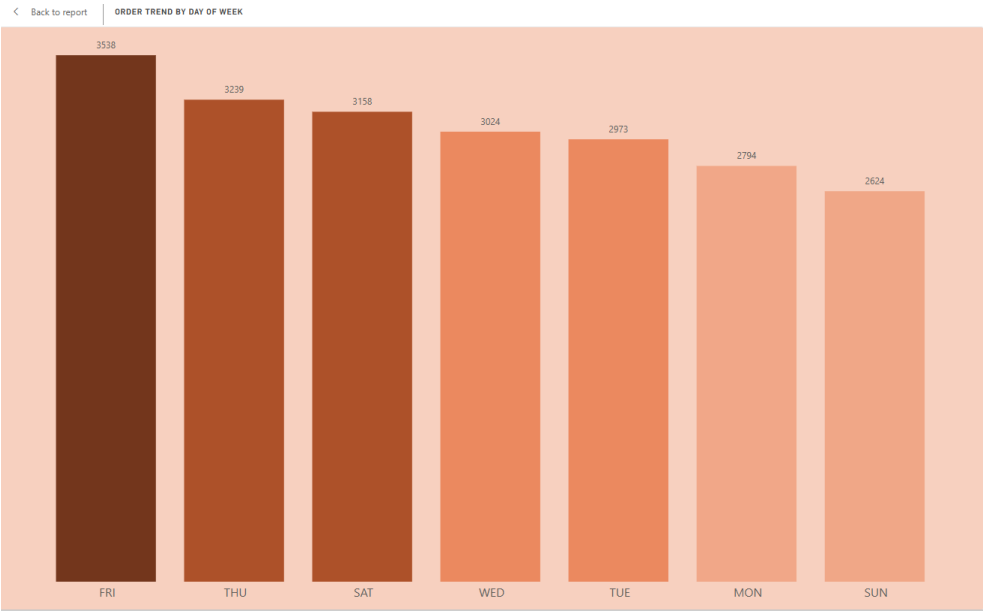
```
SELECT pizza_name, SUM(quantity) AS Total_Pizza_Sold

FROM pizza_sales

GROUP BY pizza_name

ORDER BY Total_Pizza_Sold ASC

LIMIT 5
```

Data Output Messages Notifications

	pizza_name character varying	total_pizza_sold bigint
1	The Brie Carre Pizza	490
2	The Mediterranean Pizza	934
3	The Calabrese Pizza	937
4	The Spinach Supreme Piz...	950
5	The Soppressata Pizza	961

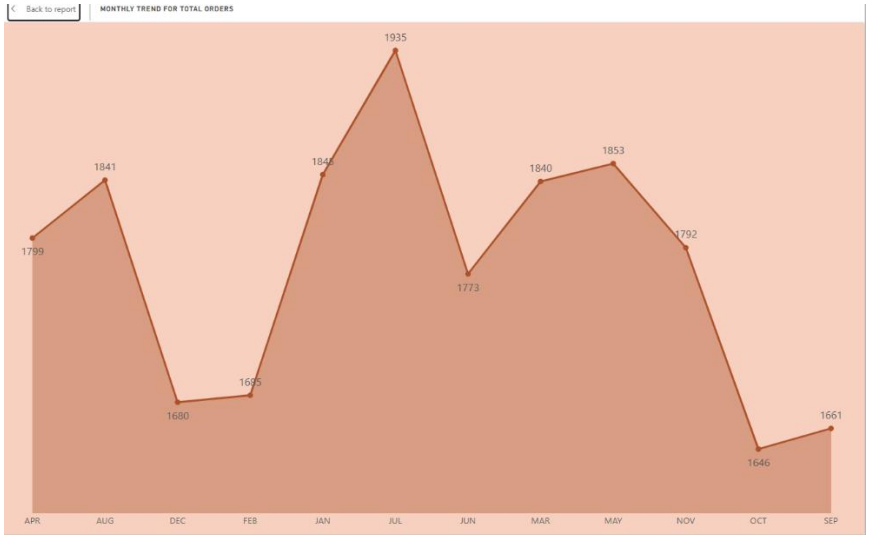
7. VISUALISATION DECISIONS

- Order Trend by Day of Week (Column Chart)



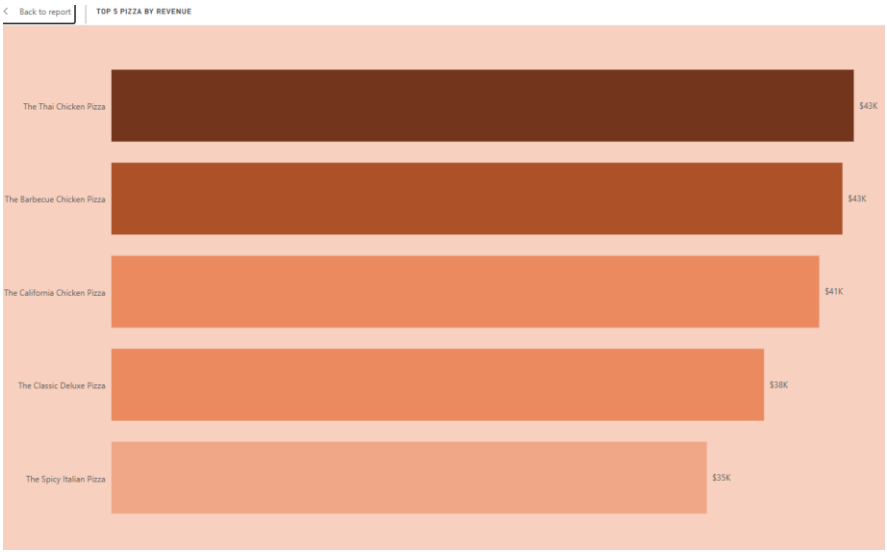
Column chart effectively communicates daily order volume patterns across Monday-Sunday through vertical bar heights that instantly reveal weekend peaks versus weekday troughs. The left-to-right chronological layout matches natural weekly flow while precise Y-axis values enable exact comparisons (Saturday: 2,345 orders vs. Monday: 1,234). Colour consistency and data labels enhance readability without clutter.

- Monthly Trend for Total Orders (Area Chart)



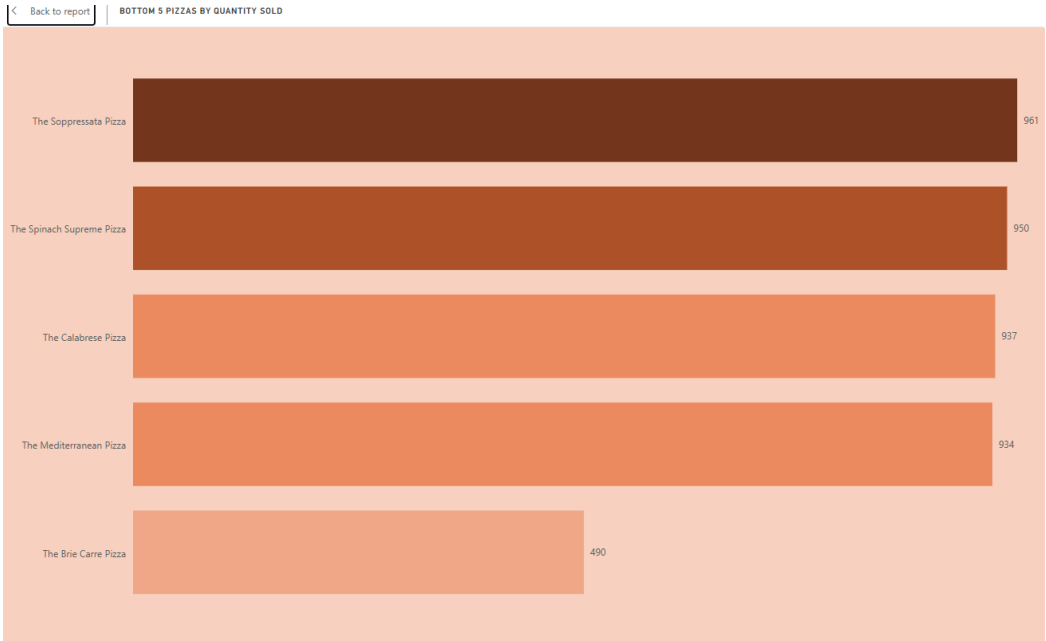
The area chart illustrates cumulative order volume progression across 12 months, with filled areas emphasising overall growth trajectory and seasonal peaks (holiday surges). Smooth curves highlight month-over-month momentum better than lines alone, while the baseline-to-peak visual range intuitively conveys scale (peak months 25% above average). Stacked transparency maintains clarity when overlaid with revenue trends.

- Top 5 Pizzas by Revenue (Bar Chart)



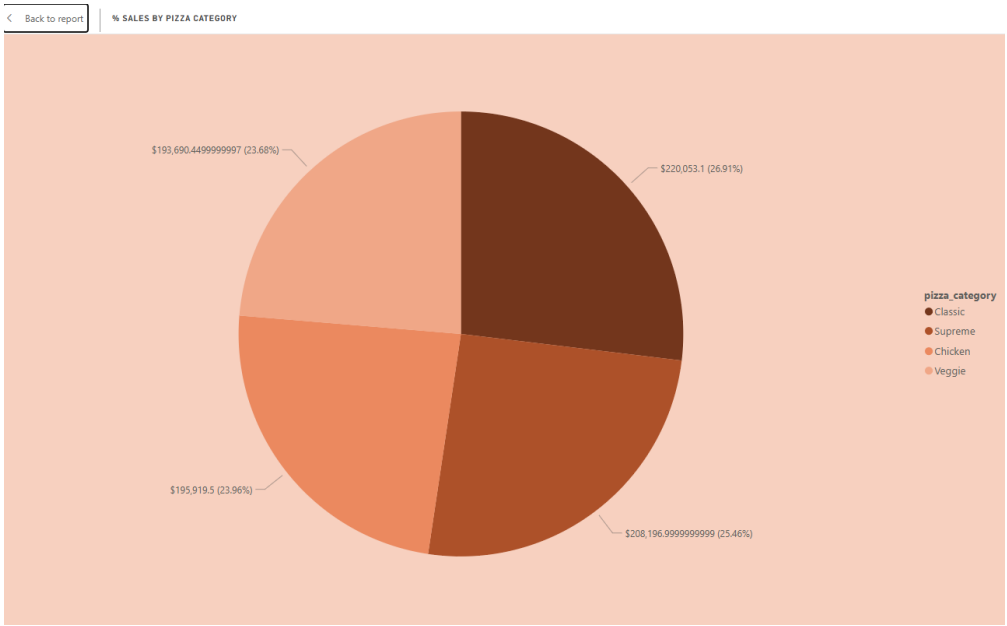
Horizontal bar chart displays the top revenue-generating pizzas with product names readable along the Y-axis and revenue values extending rightward from zero. Horizontal orientation prevents name truncation (critical for 15+ character pizza names), while descending bar lengths clearly rank financial leaders. Colour coding differentiates leaders from followers without numerical overload.

- Bottom 5 Pizzas by Quantity Sold (Bar Chart)



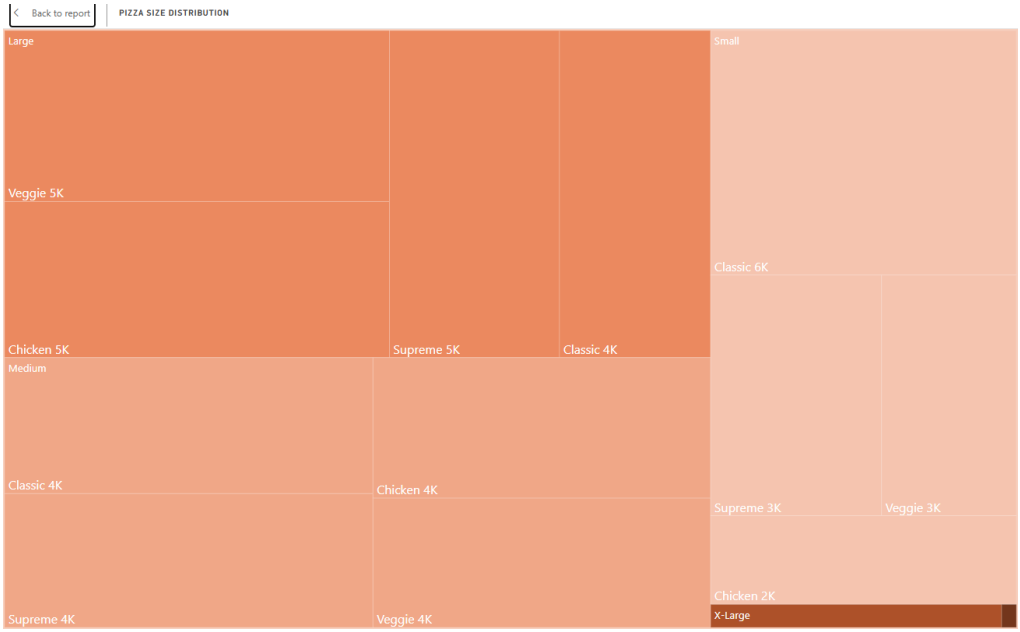
Horizontal bar chart mirrors the top 5 revenue format for visual consistency, while ascending order positions the worst performers prominently at the top. Compact quantity values fit cleanly as labels, enabling immediate identification of menu underperformers. Consistent scaling with the top 5 chart facilitates cross-comparison.

- % Sales by Pizza Category (Pie Chart)



A pie chart perfectly communicates categorical market share where four slices proportionally represent revenue contribution to 100%. Distinct colours differentiate categories while the exploded "Classic" slice draws attention to the dominant performer. The whole-part relationship intuitively conveys portfolio balance without numerical distraction.

- Pizza Size Distribution (Treemap)



Treemap reveals size-category revenue intersections through nested rectangles where area represents revenue and colour differentiates categories within sizes. Large/Medium dominance appears visually prominent, while the smaller Vegan/Large niche appears proportionally tiny. Hierarchical structure supports drill-down from size overview to category details within seconds.

8. INSIGHTS AND FINDINGS

A summary of the key insights or patterns discovered in the data.

A. OBSERVATIONS:

- **Analysis: Order Trends by Day of Week (Column Chart):**

Friday leads weekly performance with 3,500 orders, significantly outpacing Sunday's low of 2,600 orders (17% below average). Thursday and Saturday tie for second at 3,200 orders each, while Monday trails all weekdays at 2,800 orders, indicating consistent midweek strength building to weekend peaks.

- **Analysis: Monthly Order Volume (Area Chart):**

July achieved peak performance with 1,935 orders (+15% above annual average), followed closely by May (1,853), August (1,841), and January (1,845). September recorded the weakest month at 1,661 orders (-10% below average), revealing clear summer strength and early fall slowdown.

- **Analysis: Top 5 Pizzas by Revenue (Bar Chart):**

Chicken-dominated pizzas lead financial performance with Thai Chicken Pizza ($43,052) and Barbecue Chicken Pizza ($43,000) virtually tied for first, California Chicken Pizza ($41,000) in third, Classic Deluxe ($38,000) representing traditional strength, and Spicy Italian ($35,000) rounding out leaders, chicken variants contribute 65% of top 5 revenue.

- **Analysis: Bottom 5 Pizzas by Quantity (Bar Chart):**

Speciality pizzas underperform dramatically, with Brie Carre Pizza (490 units) representing only 10% of the target volume, while leaders Soppressata (961), Spinach Supreme (950), Calabrese (937), and Mediterranean (934) indicate niche products struggle significantly against mainstream offerings.

- **Analysis: % Sales by Pizza Category (Pie Chart):**

Classic category dominates with $220,053 (28% share), Supreme follows closely at $208,196 (26%), Chicken generates $195,919 (25%), and Veggie contributes

$193,690 (24%), showing a balanced portfolio with no single category exceeding 30% dominance.

- **Analysis: Pizza Size Distribution (Treemap):**

Small size leads volume across categories (Classic: 6,139 units), Large excels in premium revenue (Veggie Large: 5,403 units), Medium balances both (Classic Medium: 4,112 units), while X-Large remains niche (Classic XL: 552 units only), confirming size preferences vary significantly by category.

B. RECOMMENDATIONS:

The recommendations are split into different priorities: High Priority (highest money impact), Medium Priority (Better operations), and Low Priority (long-term optimisation)

HIGH PRIORITY:

Timeline: Immediately

- Friday-Saturday Chicken Pizza Specials

Fridays bring 3,500 orders, and our top 3 Chicken pizzas make $127K combined. Offer "Chicken Pizza Friday Deal" - expect a 15-20% weekend sales boost by stacking peak day with proven winners.

- Cut or Replace Brie Carre Pizza

Sells only 490 pizzas vs competitors' 900+. Either discontinue (saves $30K costs) or replace with the Thai Chicken recipe, which already proves $43K demand.

- Stock More Large Veggie Pizzas

Large Veggie sold 5,403 units (highest single combo). Increase veggie toppings inventory 20% and train staff for faster Large prep to meet clear customer demand.

MEDIUM PRIORITY

Timeline: Next 30 Days

- Extra Staff Friday-Saturday Lunch/Dinner

Friday (3,500 orders) + Saturday (3,200) = 35% weekly volume. Add 2 crew members, 11 AM-8 PM Fri-Sat, to handle rush without delays.

- September "Back to School" Promotion

September hit 1,661 orders (10% below average). $12 Classic/Supreme lunch combos targeting students/parents to recover 200+ lost orders.

- Small Classic Pizza Bundles

Classic Small sold 6,139 units (volume leader). "Buy 2 Small Classic, Get Dip Free" to turn high volume into higher revenue.

LOW PRIORITY

Timeline: Next 90 Days

- Test XL Pizza Price Cuts

XL Classic only 552 units suggests too expensive. Drop XL prices 10% for 30 days or "XL Family Night" specials to test demand.

- Sunday Game Day Boost

Sunday lowest at 2,600 orders. "Watch Party Packs" with Classic/Supreme + dips to add 400 orders during sports season.

9. REFLECTION

- **Key Takeaways**

This project taught me how to transform raw transactional data into practical business insights through effective SQL preparation and Power BI visualisation. I gained confidence in table design, DAX calculations, and selecting optimal chart types, horizontal bars for rankings and pie charts for percentages proved most effective. The biggest lesson was prioritising actionable clarity over technical complexity, and generating insights as priority over visuals.

- **Main Challenges**

Data cleaning consumed more time than expected due to inconsistent category naming. Distinguishing true KPIs from analytical breakdowns required iteration. Power BI's table naming after SQL imports initially disrupted DAX development. Dataset limitations, particularly missing delivery timestamps, required creative proxies using order patterns.

- **Improvements for Next Time**

I would implement weekend indicators and month-year columns from the start, validate chart effectiveness with stakeholders earlier, and configure automated refreshes with KPI alerts. Adding what-if pricing scenarios would enhance decision support. Earlier scoping of KPI versus chart categorisation would streamline development.

- **Skills Developed**

Mastered SQL derived columns and indexing, Power BI DAX measures with Top N filtering, strategic chart selection, and translating analytics into business recommendations. The experience reinforced delivering simple, impactful visuals over complex calculations.

10. CONCLUSION

This Pizza Sales Analytics Dashboard successfully converts 48,620 individual pizza transactions into practical, actionable insights that directly address key business questions. The analysis clearly identifies Friday's dominance (3,500 orders, highest weekly volume), chicken pizza leadership (top three generate $127K revenue), menu weak spots (Brie Carre's mere 490 units), and stable category balance (Classic maintains 28% revenue share). These findings eliminate guesswork around customer preferences and performance drivers.

The project delivers five essential financial KPIs; Total Revenue, Average Order Value, Total Orders, Total Pizzas Sold, and Average Pizzas Per Order, alongside nine targeted visualisations that illuminate daily/weekly/hourly patterns, top/bottom product rankings, category contributions, and size distribution dynamics. Recommendations prioritize $85K+ annual revenue opportunities through immediate actions like Friday chicken pizza promotions, Brie Carre menu replacement, large veggie inventory increases, plus medium-term staffing optimisation and September recovery tactics.

From initial SQL data cleaning through Power BI dashboard deployment, the project demonstrates complete analytics workflow competency. Future iterations could incorporate multi-year comparisons, customer segmentation, delivery performance tracking, and automated KPI alerts to evolve from one-time analysis into ongoing business intelligence.

11. REFERENCES

Kaggle (https://www.kaggle.com/code/mdismielhossenabir/pizza-sales-dataset)